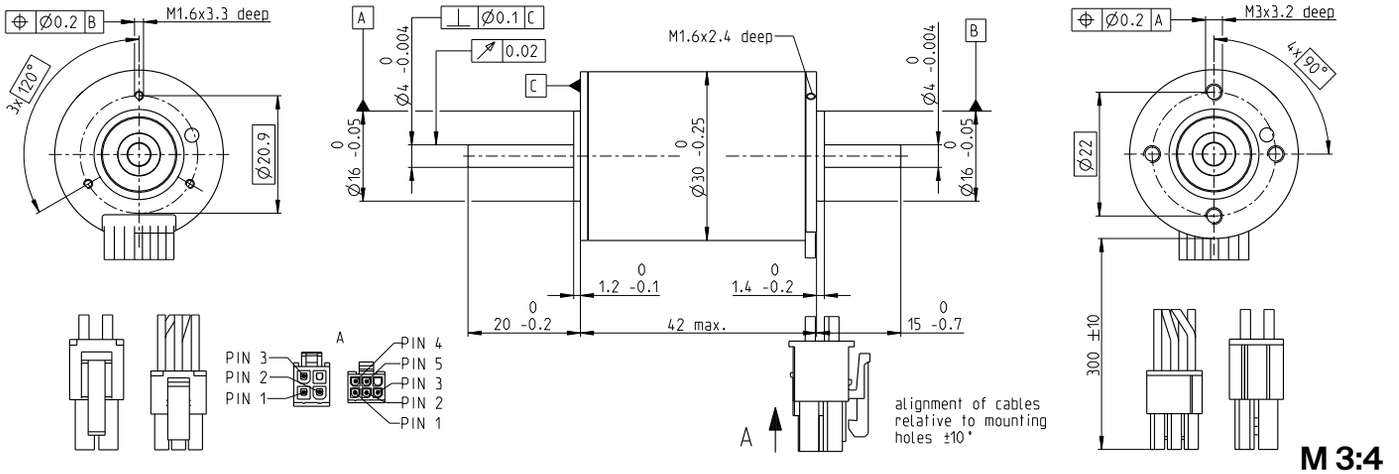
EC-i 30 ∅30 mm, brushless, 45 watt

High Torque



EC-i

Stock program
Standard program
Special program (on request)

Part numbers

with Hall sensors	539480	539481	539482	539483	539484
Motor data (provisional)					
Values at nominal voltage					
1 Nominal voltage V	12	18	24	36	48
2 No load speed rpm	8240	8250	8510	8250	8510
3 No load current mA	285	190	149	95.2	74.5
4 Nominal speed rpm	6730	6780	7050	6810	7070
5 Nominal torque mNm	64.9	67.1	63.3	67.1	63.2
6 Nominal current (max. continuous current) A	4.49	3.08	2.27	1.54	1.13
7 Stall torque mNm	469	497	491	507	496
8 Stall current A	54	41.4	31	21.8	15.9
9 Max. efficiency %	85.7	86.6	86.4	87	86.5
Characteristics					
10 Terminal resistance phase to phase Ω	0.222	0.435	0.775	1.65	3.01
11 Terminal inductance phase to phase mH	0.199	0.449	0.749	1.8	3
12 Torque constant mNm/A	13.7	20.6	26.6	41.2	53.2
13 Speed constant rpm/V	696	464	359	232	180
14 Speed/torque gradient rpm/mNm	11.4	9.94	10.6	9.43	10.3
15 Mechanical time constant ms	0.956	0.83	0.888	0.787	0.862
16 Rotor inertia gcm²	8.1	8.1	8.1	8.1	8.1

Specifications

Thermal data

17	Thermal resistance housing-ambient	11.1 K/W
18	Thermal resistance winding-housing	3.75 K/W
19	Thermal time constant winding	27.8 s
20	Thermal time constant motor	866 s
21	Ambient temperature	-40...+100°C
22	Max. winding temperature	+155°C

Mechanical data (preloaded ball bearings)

23	Max. speed		10 000 rpm
24	Axial play at axial load	< 9.0 N	0 mm
		> 9.0 N	0.14 mm
25	Radial play		preloaded
26	Max. axial load (dynamic)		5 N
27	Max. force for press fits (static)		98 N
	(static, shaft supported)		2000 N
28	Max. radial load, 5 mm from flange		25 N

Other specifications

29	Number of pole pairs	4
30	Number of phases	3
31	Weight of motor	156 g

Values listed in the table are nominal.

Connection motor (Cable AWG 20)

red	Motor winding 1	Pin 1
black	Motor winding 2	Pin 2
white	Motor winding 3	Pin 3
	N.C.	Pin 4

Connector	Article number
Molex	39-01-2040

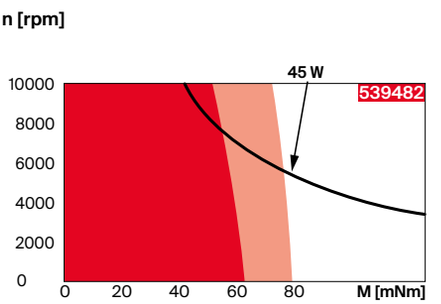
Connection sensors (Cable AWG 26)

yellow	Hall sensor 1	Pin 1
brown	Hall sensor 2	Pin 2
grey	Hall sensor 3	Pin 3
blue	GND	Pin 4
green	V_{Hall} 4.5...24 VDC	Pin 5
	N.C.	Pin 6

Connector	Article number
Molex	430-25-0600

Wiring diagram for Hall sensors see p. 69

Operating range



Continuous operation
Continuous operation with reduced thermal resistance R_{th2} 50%
Intermittent operation

Assigned power rating

Comments

Modular system

Details on catalog page 52

Gear
430_GP 32 C
474-482_GP 32 S

Sensor
509_ENX 16 EASY
510_ENX 16 EASY XT
511_ENX 16 EASY Absolute
512_ENX 16 EASY Absolute XT
518_ENX 22 EMT
519_ENX 16 RIO

Motor Control
547_DEC Module 50/5
551_ESCON 36/3 EC
551_ESCON Module 50/4 EC-S
551_ESCON Module 50/5
553_ESCON 50/5
557_ESCON2 Micro 60/5
563_EPOS4 Micro 24/5
564_EPOS4 Module 50/5
565_EPOS4 Compact 24/5 3-axes
567_EPOS4 Compact 50/5
569_EPOS4 50/5
570_EPOS4 Disk 60/8
571_EPOS4 Disk 60/12

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